

# The Economic Value of Knowledge Geometry: Topological Opportunities and Breakthrough Innovation in Scientific Networks

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## Abstract

This paper studies whether the geometry, topology, and local spectra of scientific knowledge networks contain ex ante information about breakthrough innovation. Using the OGBN-MAG subset of the Microsoft Academic Graph, I construct a dynamic scientific knowledge complex from paper-topic assignments, citation links, authorship, venues, and publication years. The central topological object is boundary-completion potential: whether a paper’s topic set spans triangular boundaries already present in the prior topic graph. I also compute local clique-complex Betti numbers and Laplacian spectral metrics on each paper’s prior topic-induced graph. Papers published between 2011 and 2016 are linked to their three-year forward citations and classified as breakthrough papers when they fall in the top five percent of their publication-year citation distribution. The preliminary evidence shows that boundary-completion potential and local spectral connectivity predict future scientific impact after controlling for authorship, topic breadth, references, year fixed effects, and venue-group fixed effects. The results suggest that breakthrough science is not only a function of research scale or centrality, but also of the high-order structure and diffusion geometry of the knowledge space in which researchers search.

*Keywords:* science of science, innovation, topology, spectral graph theory, Microsoft Academic Graph, OGBN-MAG

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## 1. Introduction

Where do breakthrough innovations come from? A large literature in economics, management, and the science of science emphasizes cumulative

knowledge, recombination, collaboration networks, and the uneven distribution of scientific opportunity (Uzzi et al., 2013; Fortunato et al., 2018; Acemoglu et al., 2016). Yet most empirical measures of scientific opportunity remain low-order: publication counts, citation counts, team size, field fixed effects, or pairwise network centrality. These measures are informative, but they do not directly characterize the high-order geometry of the knowledge space in which research ideas are combined.

This paper develops and tests a simple claim: scientific breakthroughs are more likely when researchers occupy topological opportunities in the prior knowledge space. A topological opportunity is a location where existing topics are near enough to be meaningfully connected, but the higher-order knowledge combination has not yet been stabilized. In such locations, a paper can add more than another node to the scientific graph. It can close a gap, fill a boundary, or create a bridge between previously separated knowledge regions.

The empirical setting is the OGBN-MAG subset of the Microsoft Academic Graph (Sinha et al., 2015; Hu et al., 2020). The data contain 736,389 papers, 1,134,649 authors, 8,740 institutions, 59,965 fields of study, 5.4 million citation edges, and 7.5 million paper-topic edges. For each paper, I observe publication year, venue label, authorship, references, and field-of-study assignments. I build a dynamic knowledge complex year by year. Before each paper is published, I compute whether its topics connect previously unobserved pairs, whether they span triangular boundaries in the prior topic graph, and whether they lie on negatively curved topic-pair bridges.

The core analysis uses papers published from 2011 through 2016, which provides a three-year forward citation window for all papers in the analytic sample of 485,819 papers. The outcome is either log three-year forward citations or an indicator for being in the top five percent of the publication-year citation distribution. The main specifications include controls for authors, topic breadth, references, year fixed effects, and venue-group fixed effects.

Out-of-time prediction uses nonlinear classifiers trained on 2011–2014 papers and tested on 2015–2016 papers.

The results are preliminary but encouraging in a specific sense. Regression estimates show that the boundary-completion index is positively associated with future scientific impact. Component specifications indicate that novel topic-pair formation is more closely related to tail breakthroughs than to average citations, while local Laplacian connectivity and spectral entropy are positively associated with both outcomes. Local  $\beta_1$  holes are rare in this paper-level construction and do not carry independent explanatory power. Out-of-time prediction is conservative but positive: the topology-augmented model improves ROC AUC from 0.863 to 0.864 and average precision from 0.336 to 0.337. Thus, topology and spectral graph theory are not presented here as large black-box forecasting gains. Their value is interpretive: they identify high-order and diffusion-relevant structures of scientific opportunity that are invisible in scale-based measures.

The paper contributes to three literatures. First, it adds to economics of innovation by proposing an ex ante measure of opportunity in scientific search. Second, it contributes to science-of-science research on recombination and breakthrough discovery by moving from pairwise novelty to high-order topology. Third, it connects topological data analysis and spectral graph theory (Edelsbrunner et al., 2002; Zomorodian and Carlsson, 2005; Chung, 1997; Forman, 2003) to an empirical setting where the objects have economic meaning: scientific papers, authors, fields, and future impact.

## 2. Related Works

This paper is related first to the economics of innovation and knowledge recombination. A central idea in this literature is that new ideas are produced by combining existing ideas, but that not all combinations have the same value. Uzzi et al. (2013) show that atypical scientific combinations are associated with high impact, while Acemoglu et al. (2016) study how the

network structure of technological fields shapes the direction of innovation. This paper follows that tradition but shifts the object of measurement from pairwise novelty and network position to the geometry and topology of the prior knowledge space. The key claim is not only that unusual combinations matter, but that some combinations occupy identifiable high-order structural gaps.

The paper also contributes to the science-of-science literature. Fortunato et al. (2018) synthesize evidence that scientific production is shaped by teams, institutions, careers, fields, and networks. Large-scale bibliographic graphs such as Microsoft Academic Graph and OGBN-MAG make it possible to measure these structures at scale (Sinha et al., 2015; Hu et al., 2020). Existing work often represents science as citation, collaboration, or topic networks. I use the same graph-based view of science, but construct a dynamic topic complex in which papers add edges, simplices, and local spectral structure to the prior knowledge space.

Finally, the paper connects innovation measurement to topological data analysis, graph geometry, and spectral graph theory. Persistent homology provides tools for measuring connected components, cycles, and higher-order cavities in filtered spaces (Edelsbrunner et al., 2002; Zomorodian and Carlsson, 2005). Spectral graph theory studies how Laplacian eigenvalues summarize connectivity and diffusion in graphs (Chung, 1997). Discrete curvature offers another way to identify bridge-like or high-load regions of a graph (Forman, 2003). This paper uses these tools as empirical measurements of scientific opportunity: Betti numbers and persistence diagrams describe knowledge holes, Laplacian spectra describe local diffusion geometry, and Forman curvature describes bridge-like topic pairs.

### 3. Conceptual Framework

Let  $G_t = (V_t, E_t)$  be the cumulative topic co-occurrence graph before year  $t$ . A node is a field of study and an edge indicates that two fields have

appeared together in at least one prior paper. A paper  $i$  published in year  $t$  has a topic set  $S_i \subset V_t$ . The paper can be interpreted as adding local structure to the prior knowledge graph.

The first component of opportunity is pairwise novelty:

$$N_i = \frac{1}{\binom{|S_i|}{2}} \sum_{\{u,v\} \subset S_i} \mathbf{1}\{(u,v) \notin E_t\}. \quad (1)$$

This measure captures whether the paper combines topics that had not previously appeared together.

The second component is boundary-completion potential. For triples of topics, define:

$$B_i = \frac{1}{\binom{|S_i|}{3}} \sum_{\{u,v,w\} \subset S_i} \mathbf{1}\{(u,v), (u,w), (v,w) \in E_t\}. \quad (2)$$

When the three boundary edges already exist, the new paper supplies a higher-order simplex over a previously established triangular boundary. This is a local, computable analogue of filling a one-dimensional hole in a knowledge complex.

I also compute local Betti numbers on the clique complex induced by  $S_i$  in the prior graph. Let  $C_k(S_i)$  be the vector space over  $\mathbb{F}_2$  generated by  $k$ -simplices in the local clique complex, and let  $\partial_k : C_k \rightarrow C_{k-1}$  be the boundary operator. The local Betti numbers are:

$$\beta_k(S_i) = \dim \ker(\partial_k) - \dim \operatorname{im}(\partial_{k+1}). \quad (3)$$

In the implementation,  $k = 0, 1, 2$  are computed exactly on each paper's local topic complex. This gives a small-scale TDA object rather than a verbal topology metaphor.

The spectral component uses the graph Laplacian of the same prior topic-induced graph. Let  $A_i$  be the local adjacency matrix,  $D_i$  its degree matrix,

and:

$$L_i = D_i - A_i. \quad (4)$$

I compute the second-smallest eigenvalue  $\lambda_2(L_i)$ , the adjacency spectral radius, and Laplacian spectral entropy.  $\lambda_2(L_i)$  measures local algebraic connectivity; spectral entropy captures how dispersed the local diffusion modes are.

The final graph-geometric bridge measure is based on Forman curvature. For an unweighted edge  $(u, v)$  in the prior topic graph, a simple Forman-Ricci curvature proxy is:

$$F(u, v) = 4 - d(u) - d(v), \quad (5)$$

where  $d(\cdot)$  is prior topic degree. Highly negative values identify edges that connect high-degree regions and therefore act as bridge-like, high-load parts of the knowledge geometry. For paper  $i$ , I average negative curvature over its already existing topic pairs.

The main topological opportunity index is standardized boundary closure:

$$TO_i = z(B_i). \quad (6)$$

Pairwise novelty, local Betti numbers, Laplacian spectral metrics, and negative Forman curvature are retained as decomposition variables. The object is intentionally ex ante: all components are computed using only knowledge observed before the paper’s publication year.

#### 4. Data

The analysis uses OGBN-MAG, a heterogeneous academic network extracted from Microsoft Academic Graph and distributed by the Open Graph Benchmark. The graph contains four node types: papers, authors, institu-

tions, and fields of study. It also contains four directed relation types: authors are affiliated with institutions, authors write papers, papers cite papers, and papers have fields of study.

Table 1: Dataset and analytic sample

| Quantity           | Count     |
|--------------------|-----------|
| Papers             | 736,389   |
| Authors            | 1,134,649 |
| Institutions       | 8,740     |
| Fields of study    | 59,965    |
| Citation edges     | 5,416,271 |
| Paper-topic edges  | 7,505,078 |
| Author-paper edges | 7,145,660 |
| Analytic papers    | 485,819   |

The paper-level outcome is three-year forward citations. For a paper  $i$  published in year  $t$ , I count citations from papers published in years  $t + 1$ ,  $t + 2$ , and  $t + 3$ . The breakthrough indicator equals one if the paper is in the top five percent of the forward-citation distribution among papers published in the same year.

## 5. Empirical Strategy

The baseline regression is:

$$Y_{i,t+3} = \alpha + \beta TO_{i,t} + X'_{i,t}\gamma + \delta_t + \mu_{v(i)} + \varepsilon_i, \quad (7)$$

where  $Y_{i,t+3}$  is either log three-year forward citations or the breakthrough indicator.  $TO_{i,t}$  is the topological opportunity index.  $X_{i,t}$  includes log authors, log topic count, and log references.  $\delta_t$  denotes publication-year fixed effects.  $\mu_{v(i)}$  denotes venue-group fixed effects for the largest venue groups and an omitted group for all remaining venues.

Table 2: Summary statistics

| Variable                      | Mean     | Std. dev. | P25      | Median   | P75       |
|-------------------------------|----------|-----------|----------|----------|-----------|
| Three-year forward citations  | 4.885    | 11.639    | 1.000    | 2.000    | 5.000     |
| Breakthrough indicator        | 0.051    | 0.221     | 0.000    | 0.000    | 0.000     |
| Novel topic-pair share        | 0.127    | 0.152     | 0.000    | 0.067    | 0.200     |
| Boundary-completion potential | 0.725    | 0.273     | 0.500    | 0.800    | 1.000     |
| Negative Forman curvature     | 7444.968 | 4242.967  | 3693.286 | 6954.364 | 10835.667 |
| Topological opportunity index | 0.000    | 1.000     | -0.827   | 0.274    | 1.008     |
| Local $\beta_0$               | 1.056    | 0.276     | 1.000    | 1.000    | 1.000     |
| Local $\beta_1$               | 0.001    | 0.033     | 0.000    | 0.000    | 0.000     |
| Local $\beta_2$               | 0.000    | 0.002     | 0.000    | 0.000    | 0.000     |
| Local $\lambda_2(L)$          | 3.865    | 1.913     | 2.000    | 4.000    | 6.000     |
| Local spectral radius         | 4.453    | 0.643     | 4.162    | 4.702    | 5.000     |
| Local spectral entropy        | 1.556    | 0.121     | 1.554    | 1.598    | 1.609     |
| Authors                       | 8.895    | 91.279    | 2.000    | 4.000    | 6.000     |
| Topics                        | 10.319   | 1.397     | 10.000   | 11.000   | 11.000    |
| References                    | 6.212    | 8.791     | 1.000    | 3.000    | 8.000     |

The coefficient  $\beta$  should not be interpreted as a fully causal estimate in this draft. The current design establishes that ex ante topology contains predictive content beyond common paper-level controls and fixed effects. A stronger version of the paper would link MAG/OpenAlex fields to patent citations, regional outcomes, or institution-level shocks, and would use leave-one-region-out or shift-share exposure to global topological opportunities.

## 6. Results

Figure 1 shows the evolution of the knowledge complex. Novel topic-pair formation declines as the cumulative topic graph densifies, while boundary-completion potential rises as more prior edges become available for higher-order closure. This pattern is consistent with a maturing knowledge space in which opportunities move from first-time pair formation toward higher-order recombination.

Figure 2 provides a dedicated TDA visualization for an active frontier band of the topic complex: fields ranked 101–180 by frequency. The most fre-



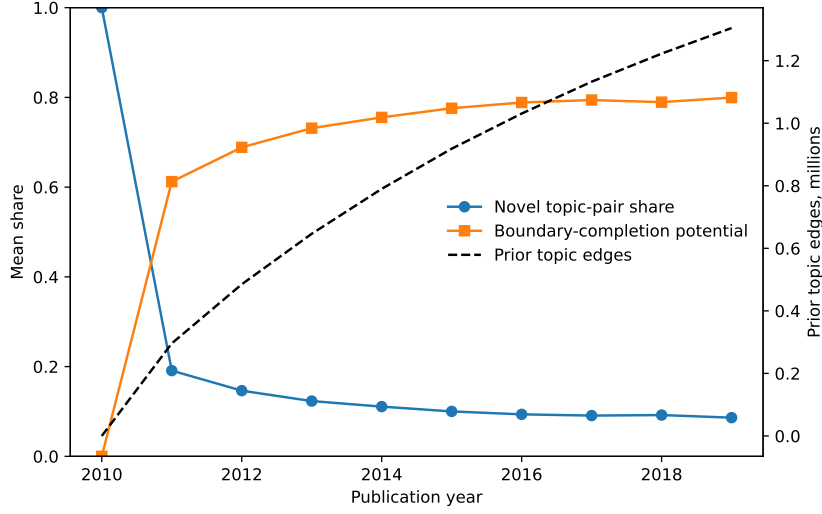


Figure 1: Evolution of topic-pair novelty and boundary-completion potential

quent core fields are already saturated, so this second tier is more informative about unresolved structure. The filtration runs from strong to weak knowledge ties: the highest-frequency topic-pair edges enter first, weaker bridges enter later, and triangles enter when genuine three-topic co-occurrences provide higher-order closure. The Betti curves summarize how components merge and one-dimensional holes appear as weaker bridges are admitted, while the persistence diagram and barcode show the lifetimes of the longest-lived knowledge holes.

Figure 3 shows the paper-level local homology and spectral graph measures. Local  $\beta_1$  is sparse, which is informative: in this paper-level construction, the prior topic neighborhoods are usually connected and triangle-rich. By contrast, local algebraic connectivity and spectral entropy vary systematically over time, giving usable variation for spectral graph analysis.

Figure 4 bins papers by the topological opportunity index. Papers in higher-opportunity bins receive more future citations and are more likely to become breakthrough papers. The relationship is not a proof of causality, but it is the first empirical fact the paper seeks to explain.

Persistent homology of field-frequency ranks 101-180

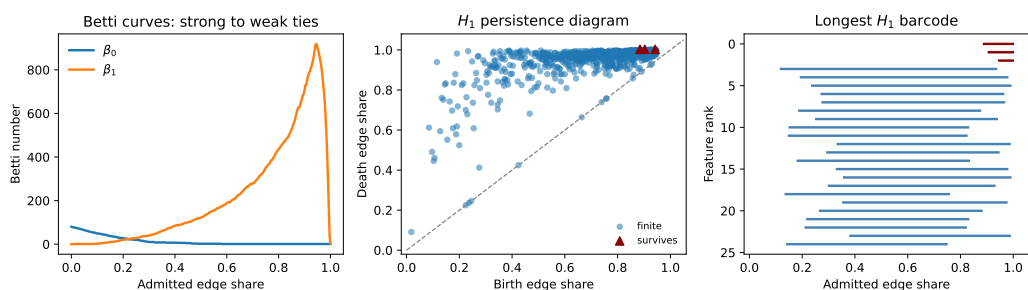


Figure 2: Persistent homology of the scientific knowledge complex

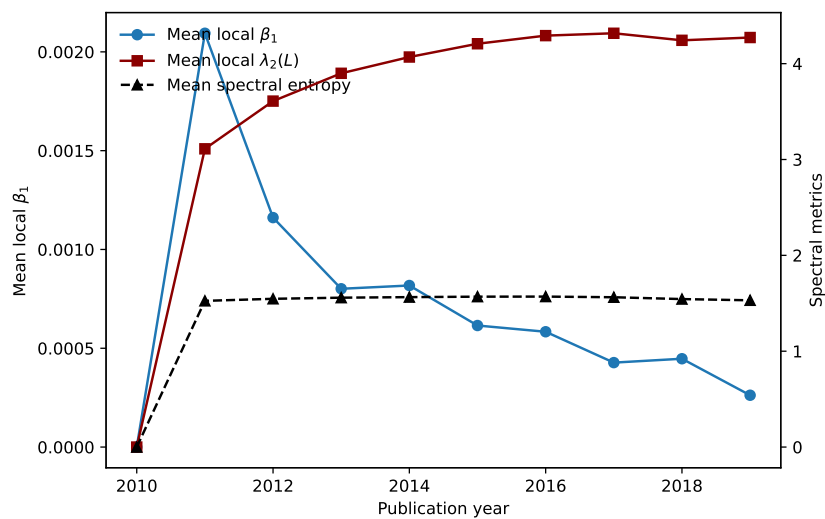


Figure 3: Local TDA and spectral graph measures

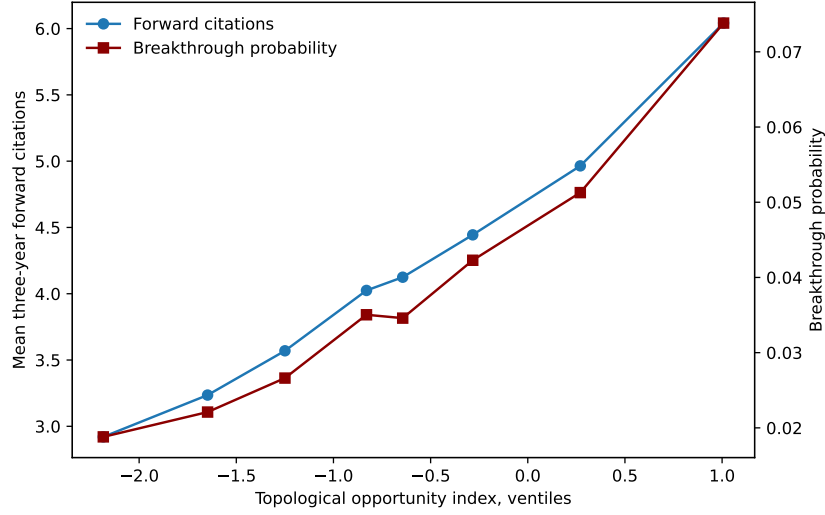


Figure 4: Topological opportunity and future scientific impact

Table 3: Topology, geometry, and future scientific impact

|                    | (1)       | (2)                 | (3)                  | (4)     | (5)                 | (6)                  |
|--------------------|-----------|---------------------|----------------------|---------|---------------------|----------------------|
| TO index           |           | 0.029***<br>(0.001) |                      |         | 0.002***<br>(0.000) |                      |
| Novel pairs        |           |                     | -0.014***<br>(0.003) |         |                     | 0.004***<br>(0.001)  |
| $\beta_1$ holes    |           |                     | -0.000<br>(0.001)    |         |                     | -0.000<br>(0.000)    |
| $\lambda_2(L)$     |           |                     | 0.008***<br>(0.003)  |         |                     | 0.006***<br>(0.001)  |
| Spectral entropy   |           |                     | 0.012***<br>(0.002)  |         |                     | 0.001**<br>(0.000)   |
| Negative curvature |           |                     | -0.013***<br>(0.001) |         |                     | -0.001***<br>(0.000) |
| Outcome            | Log cites | Log cites           | Log cites            | Top 5%  | Top 5%              | Top 5%               |
| Specification      | Base      | + TO                | + TDA/Spec.          | Base    | + TO                | + TDA/Spec.          |
| Controls           | Yes       | Yes                 | Yes                  | Yes     | Yes                 | Yes                  |
| Year FE            | Yes       | Yes                 | Yes                  | Yes     | Yes                 | Yes                  |
| Venue-group FE     | Yes       | Yes                 | Yes                  | Yes     | Yes                 | Yes                  |
| $N$                | 485,819   | 485,819             | 485,819              | 485,819 | 485,819             | 485,819              |
| $R^2$              | 0.309     | 0.310               | 0.310                | 0.143   | 0.143               | 0.143                |

Notes: HC1 robust standard errors in parentheses. The analytic sample contains papers published from 2011 through 2016, allowing a three-year forward citation window. Topology variables are standardized within the analytic sample. \* $p < 0.10$ , \*\* $p < 0.05$ , \*\*\* $p < 0.01$ .

Table 3 reports the main regression evidence. The boundary-completion index is positively associated with both log forward citations and the breakthrough indicator. Component specifications show a useful distinction: first-time topic-pair novelty is negatively associated with average citations but positively associated with the probability of reaching the top tail. Local  $\lambda_2(L)$  and spectral entropy are positive in both outcome models, consistent with the interpretation that well-connected local knowledge neighborhoods diffuse more effectively. Local  $\beta_1$  holes are too sparse to explain much variation in this draft.

Table 4: Out-of-time prediction of breakthrough papers

| Model               | ROC AUC | Average precision | Test $N$ |
|---------------------|---------|-------------------|----------|
| Baseline controls   | 0.863   | 0.336             | 156,775  |
| Controls + topology | 0.864   | 0.337             | 156,775  |

Notes: Histogram gradient-boosted classifiers train on 2011–2014 papers and test on 2015–2016 papers. Baseline controls include log authors, log topics, log references, and publication year. The topology model adds local Betti-number and spectral graph features. Breakthrough papers are defined as top-five-percent papers within publication year by three-year forward citations.

Table 4 and Figure 5 report out-of-time prediction. Models are trained on papers from 2011–2014 and tested on 2015–2016 papers. The prediction exercise uses histogram gradient boosting so that local topology can enter through threshold effects and feature interactions rather than only as a linear main effect. Adding local Betti-number and spectral graph features produces a small but positive gain over the baseline in both ROC AUC and average precision. This should be read as an incremental forecasting result rather than a large black-box performance claim.

## 7. Discussion

The main interpretation is that scientific knowledge has economically meaningful shape. A paper’s future impact depends not only on the num-

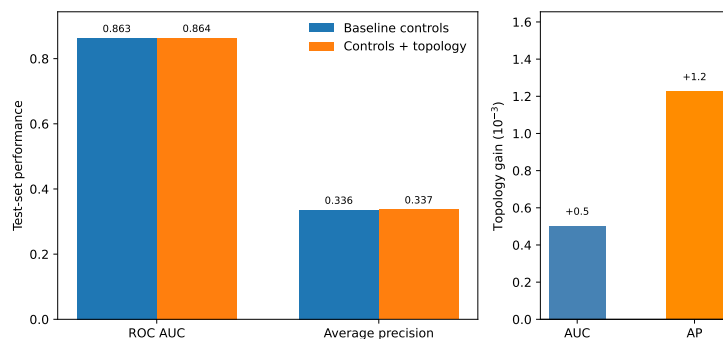


Figure 5: Out-of-time prediction performance

ber of authors, the breadth of topics, or the prestige of the venue group, but also on whether its topic combination lies in a region of the knowledge complex where prior work has created a structured opportunity. Pairwise novelty captures unexplored combinations. Boundary-completion potential captures higher-order closure. Local Betti numbers capture small-scale homology. Laplacian eigenvalues capture diffusion geometry. Negative curvature captures bridge-like locations in the topic graph.

This interpretation connects naturally to models of recombinant innovation. If researchers search locally in knowledge space, then the distribution of opportunities is shaped by prior discoveries. Some areas are saturated; others contain gaps that are visible only when the knowledge space is represented as a high-order object rather than as a list of fields or citations. Topology provides a vocabulary for those gaps.

The current draft has three important limitations. First, the outcome is scientific impact rather than direct economic value. Citations are a meaningful measure of scientific influence, but a top-journal economics version should link papers to patent citations, firm innovation, or regional outcomes. Second, the current TDA measures are exact only on local paper-level clique complexes; they are not yet persistent homology over a global filtration of the MAG graph. Third, the research design is predictive rather than causal.

Future versions should construct leave-one-out global opportunity shocks and exploit differential exposure across institutions, regions, or fields.

## 8. Conclusion

This paper proposes a new way to measure scientific opportunity. Instead of asking only how many papers, authors, citations, or topics are present, it asks where a paper lies in the geometry and topology of the prior knowledge space. Using OGBN-MAG, the preliminary evidence shows that ex ante topological opportunities predict future breakthrough science.

The broader implication is that innovation policy and science strategy may benefit from measuring the shape of knowledge. If high-value discoveries often arise from topological gaps, then institutions and funding agencies should care not only about field size or recent citation momentum, but also about the latent geometry of recombination.

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